

ARAVIND RAM

AI INFRASTRUCTURE ENGINEER

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PROFFESIONAL SUMMARY

AI Infrastructure Engineer with 2+ years building scalable Kubernetes/AWS infrastructure to host and serve AI and data workloads at scale (60 TB+, 200+ servers). Provisioned EKS platforms with Terraform and GitOps that run MCP servers and LLM-assisted services, integrated observability for AI ops, and operated the data platforms (ClickHouse, MongoDB) that feed analytics and AI pipelines. Strong infrastructure foundations applied to the emerging AI-serving stack. CKAD-certified.

SKILLS

- **AI/Compute Infrastructure** : Kubernetes (EKS) workload scheduling, **Karpenter** autoscaling, CPU Manager pinning for compute-heavy workloads, container-based hosting of MCP/LLM services
- **IaC & Platform**: Terraform, Helm, ArgoCD GitOps, Docker
- **Data Infrastructure for AI**: ClickHouse (60 TB+ analytics), MongoDB, hot/cold tiered storage (EBS→S3), **pipelines** feeding AI/analytics
- **AI Tooling Infra**: MCP servers (FastMCP, MetaMCP), Temporal workflow orchestration, LLM-assisted ops services
- **Cloud (AWS)**: IAM/IRSA, VPC, EC2, S3, EKS, Lambda, Route53, ASG, ACM, **Bedrock and SageMaker**
- **Observability**: Prometheus, Grafana, AlertManager (PromQL), **Elastic Stack** — metric correlation for AI/ops
- **Languages**: Python, Go, Bash

EXPERIENCE

FRESHWORKS, CHENNAI

2023- PRESENT

Infrastructure Engineer, AI Workloads

- **Infrastructure for AI & Data Services**
 - Provisioned and operated **EKS infrastructure** (Terraform + **ArgoCD** GitOps) hosting **MCP** servers, LLM-assisted tooling, and data services at **200+ servers / 60 TB+ scale**
 - Built **CI/CD** pipelines (**GitHub Actions**) for validation, **ECR** image sync, and automated deployment of **AI/data** services
- **Compute Scheduling & Performance**:
 - Tuned **Kubernetes CPU Manager** with dedicated **physical-CPU pinning** for **compute-intensive workloads**, reducing CPU throttling **~90%** and **improving** performance **~2x**
 - Implemented Karpenter autoscaling to scale compute elastically and cost-efficiently for large analytics/AI workloads
- **AI Tooling on the Platform**
 - **Built MCP** servers (FastMCP/MetaMCP) and an **LLM incident assistant** (**Temporal**) running on the platform — query-pattern analysis, **alert** correlation, **RCA** generation
 - Built an **AI developer-tooling** platform (**provisioning** **Cursor/Claude/Kiro**) with usage **analytics** and a **credit-wallet** metering system

- **Data Infrastructure & Storage**
 - Operated the **ClickHouse/MongoDB** data platforms **feeding analytics and AI workflows**, with cross-region DR and PITR
 - Implemented hot/cold tiered storage (**EBS → S3**) to **scale infrastructure cost-efficiently** for large datasets; improved query performance ~40%
 - **Reliability & Cost**
 - Standardized **GitOps** delivery, **reducing manual operational toil ~70%**
 - Drove **~30% AWS cost reduction** via instance **right-sizing** and **autoscaling**
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PROJECTS

- AI Tooling Platform – infra + analytics + credit-wallet metering for org-wide AI tools
- ClickHouse MCP Server – FastMCP/MetaMCP AI-assisted operations service
- SOPCTL Incident Assistant – Temporal + LLM automated RCA
- CostSenseAI – ML/LLM cloud cost-optimization platform (React, FastAPI)

EDUCATION

Bachelor of Science (BS) Degree in Data Science and Applications

Indian Institute of technology, Madras 2023 - 2027

COURSEWORK

CKAD: Certified Kubernetes Application Developer

LANGUAGES

Tamil - Native

English - Fluent